INCLUSIVE SCIENCE LESSON PLAN

LIFE CYCLE OF A PLANT

Forty minutes on the class 3D model, with real seeds on Stage 1

Prospective Teacher	Hassnain	**Class**	Six / _____
Subject	Science	**Date**	_____
Duration	40 minutes	**Class Strength**	_____
Topic	Life Cycle of a Plant	**Curriculum**	_____

The lesson teaches the five stages written on the class model: Seed, Sprout, Seedling, Adult Plant and Mature Plant. The model is the multisensory object. Real gram and bean seeds are fixed at Stage 1 so blind learners can feel a true seed, follow the white root threads, and compare leaf sizes by touch. Deaf learners take the same science from the model's colours, diamond numbers, pictures and written words, and from sign language. The three methods run together. Every child visits each one.

1. Specific Objectives

By the end of the lesson, students will be able to:

Lower-order thinking (Remembering and Understanding)

- Identify the five stages on the model by speaking, signing, pointing or touching: seed, sprout, seedling, adult plant, mature plant.
- Match one real object or model part to its stage: the dry seed to the seed, the soaked seed with a tiny root to the sprout, the small leaf to the seedling, the fuller plant to the adult plant, and the tallest plant to the mature plant.
- Name four needs of a plant and find each one on the model: sunlight on the gold sun, water on the silver clouds, soil on the orange soil panel, and air in the blue space around the leaves.

Higher-order thinking (Applying, Analysing and Creating)

- Arrange the five stages from 1 to 5 and show that seeds from the mature plant return the story to Stage 1, which is why this is called a cycle.
- Compare the seedling and the mature plant by leaves, stem and roots, and give one likeness and one difference.
- Predict what happens if a seed gets no water, or if the sun on the model is covered, and give a reason.
- Choose one stage and justify the choice with one feature: the real seed, the root threads, the stem, the leaves, or the making of new seeds.

2. Skills and Attitudes

- Main skills: observing with the eyes and with the hands, ordering stages, comparing two stages, and explaining a cause.
- Supporting skills: handling real seeds safely, reading numbers and picture cards, signing key science words, and telling a partner what was found.
- Attitudes: careful hands on the model, patience while someone explores by touch, respect for speech, sign, pointing and touch as full answers, and care for living plants.

3. Teaching Aids and Resources

One model carries the lesson. The cards and the real seeds give every learner a way in.

- **The class model:** an open 3D board. The left panel has the title Life Cycle of a Plant, a gold edge, and five labels with diamond numbers: 1 Seed, 2 Sprout, 3 Seedling, 4 Adult Plant, 5 Mature Plant. The centre is an orange-brown soil cross-section with white root threads and a green grass edge. The right side is a blue sky with a gold sun, silver clouds, and green paper plants that grow taller from seedling to mature plant.

- **Real seeds on the model:** four or five dry gram (chana) seeds and two dry beans, fixed in a small cloth pocket at label 1. The same seeds are also kept in a lidded box so a learner can hold one and then put it back. Large seeds are used so they are easy to feel and hard to lose.
- **Soaked bean:** one bean soaked overnight, kept in a small lid beside label 2. It shows the sprout: softer, larger, and often with a tiny root. It is brought on the day of the lesson and is handled only with clean hands.
- **Leaf set:** three loose paper leaves (small, medium, large) with raised glue veins: one vein, two veins, three veins. They match the leaves already glued on the model.
- **Visual cards:** diamond cards 1 to 5 in large print, each with the stage word and a simple drawing; a colour key card; four picture cards (sun, cloud, soil, air); a large curved arrow card; a sequencing mat.
- **Word and sign cards:** word cards for seed, sprout, seedling, root, stem, leaf, plant, grow, sun, water, soil, air and cycle. Urdu is printed beside the English so the class can use both.
- **Tactile numbers:** raised dots or Braille numbers 1 to 5 on the diamond labels, or a separate set of tactile number cards, so the numbers can be read by touch as well as by eye.
- **Classroom set:** whiteboard and marker, soap or sanitiser, a tray under the seed box, and wipes. The model stays on one table at a height every child can reach, with a clear path to it.

3.1 Preparing the model before the lesson

Do this before the children arrive. The painted model stays as it is. These additions make it teachable by touch as well as by sight.

- Stitch or tape a small cloth pocket onto the soil panel next to label 1. Place the real gram and bean seeds in that pocket. The black oval shapes already on the soil show where the seeds belong.
- Keep the white root threads. If they are too thin to feel, lay a thicker yarn line along the same path, from the seed pocket down into the soil and up toward the stem.
- Check that the sun’s rays are stiff enough to feel and that each cloud is a soft raised shape. The soil panel should feel rougher than the blue sky board.
- Add a raised dot or a Braille number on each diamond, 1 through 5.
- On the morning of the lesson, soak one bean and bring it in a lid. Ask families beforehand about seed allergies. Children with a known allergy use the paper model only and do not handle loose seeds.
- Agree the sign language with the deaf learner, the family, or an interpreter. In most Pakistani classes this is Pakistan Sign Language. Write down the signs you will actually use. Finger-spell any word that has no agreed sign, and keep that word card in view.



Fig. 1. The class model used in this lesson. *Left: title and stages 1 to 5. Centre: soil cross-section; real seeds are added at Stage 1 before teaching. Right: sun, clouds, and plants increasing in size. White threads are the roots.*

3.2 Key words for cards, board and signs

Teach the sign, the English word and the Urdu word with the model part, all at the same time. Confirm every sign before the lesson. Finger-spell sprout, seedling, mature and cycle if the learner has no sign for them yet. On each word card, handwrite the Urdu word in Urdu script under the Roman Urdu given here.

English	Roman Urdu	On the model	Sign note
Seed	beej	Real seeds in the pocket at 1	Confirm the learner's sign
Sprout	ankur	Root threads; soaked bean at 2	Finger-spell if needed
Seedling	nau-khez pauda	Short stem, small leaves at 3	Finger-spell if needed
Adult plant	bara pauda	Taller stem, more leaves at 4	Sign PLANT + point to 4
Mature plant	baligh pauda	Tallest plant at 5	Finger-spell MATURE
Root	jar	White threads in the soil	Confirm the learner's sign
Stem / Leaf	tana / patta	Brown stem and green leaves	Confirm both signs
Sun	suraj	Gold sun on the blue board	Confirm the learner's sign
Water	pani	Silver clouds (rain)	Confirm the learner's sign
Soil	mitti	Orange soil panel	Confirm the learner's sign
Air	hawa	Open blue space by the leaves	Confirm the learner's sign
Grow / Cycle	barhna / chakkar	Arrow from stage 5 back to the seed	Finger-spell CYCLE

4. Classroom Management and Methodology

4.1 Classroom management

- The model stays in one place for the whole lesson. Tell each blind learner, in words, where it is: title board on the left, soil in the middle, sky on the right, numbers 1 to 5 down the title side.
- The teacher faces deaf learners when signing or speaking. Write on the board first, then turn back so the face and hands stay visible. Stand so the light falls on the face, and keep hands away from the mouth.
- One child responds at a time. A correct answer may be spoken, signed, pointed, written, drawn, or shown by placing a hand on the model.
- Ask before guiding a hand. Offer the real seed in an open palm and let the learner reach for it.
- Seeds stay in the pocket or the closed box. They are felt and returned. Children wash their hands after the seed work. Glitter from the title board stays off the face and out of the eyes.
- Paths to the model are clear. Cards and the seed box sit within comfortable reach. A partner moves a piece only when the learner asks.

4.2 Teaching methodology

- Short demonstration on the real model, then guided exploration, then practice in mixed groups.
- Three access methods, written out in the activity section below: multisensory work for blind learners, and visual work together with sign language for deaf learners.
- Universal Design for Learning: the same science is offered through objects, pictures, print and signs, and children may show learning in more than one way.
- Groups of about four. Children choose roles and may swap them: Seed Keeper, Card Setter, Sign Leader, Reporter. A role is a job, and it is offered to any child who wants it.
- A child who is still learning the five names works with stages 1, 3 and 5 first. Stages 2 and 4 are added once those three are secure.

4.3 Activity Section — Multisensory, Visual and Sign Language

This is the activity section of the lesson. Three methods are planned. Each method has its own activities. The procedure in Section 5 says which activities run in the 40 minutes, so the bank below is teachable inside one period.

Who each method is for

- **Multisensory method —** Blind and visually impaired learners. The model itself is the multisensory material: real seeds, yarn roots, leaves of three sizes, a stiff sun, soft clouds and a rough soil panel. Smell of the dry seed and a simple body sequence add two more senses. Touch leads.
- **Visual method —** Deaf learners. Meaning is carried by colour, size, number, picture and large print on the model and on the cards. Speech is written down. Colour is always paired with a number or a word, so the meaning is still there for a child who does not use that colour.
- **Sign language method —** Deaf learners, alongside the visual method. The teacher or interpreter signs while pointing at the model. Hearing children learn the same signs, so signing is a class language and the deaf learner can answer in it.

Method	Planned for	What the child uses	Activities today
Multisensory	Blind and visually impaired learners	Real seeds, soaked bean, yarn roots, three leaf sizes, sun, clouds, soil texture	A1 in the demonstration A2, A3, A4 at the touch station A5 in guided practice A6 in the recap
Visual	Deaf learners	Gold sun, silver clouds, blue sky, green plants, diamond numbers, picture cards, large print	B1 in the demonstration B2, B3, B4 at the picture station B5 in the recap

Method	Planned for	What the child uses	Activities today
Sign language	Deaf learners	Pakistan Sign Language, or the child's own sign language, plus the written word	C1 in the demonstration C2 and C3 at the sign station C4 in guided practice C5 in the recap

A. Multisensory activities (for blind and visually impaired learners)

The child uses hands first. The teacher names the stage after the child has felt the object, and keeps the language short: "Stage 1. Seed. Hard. In the soil."

A1	Seed in the Pocket
When	Development, about 2 minutes, with the whole class watching or listening.
Materials	Cloth pocket of real gram and bean seeds fixed at label 1. Closed box with the same seeds.
Steps	1. Tell the learner where label 1 is, then offer the pocket. 2. The learner takes one seed, feels it, and shows or says whether it is hard, smooth, large or small. 3. The seed goes back into the pocket. The learner taps the diamond at label 1. 4. A second child repeats with the other kind of seed, so the class meets both gram and bean.
Look for	The learner finds a real seed and links it to Stage 1 without being handed the answer first.

A2	Dry Seed, Soaked Seed
When	Touch station in the group activity.
Materials	One dry bean and one soaked bean in a lid. Card or spoken frame: "This one had water."
Steps	1. The learner feels the dry bean, then the soaked bean. 2. Ask which one is bigger and which one is softer. 3. If a tiny root is present, the finger rests on it. Say: "Water woke the seed. The root comes out first. This is the sprout, Stage 2." 4. The learner places the soaked-bean lid beside label 2.
Look for	The learner can tell the dry seed from the soaked seed and can place the soaked seed at Stage 2.

A3	Follow the White Root
When	Touch station, just after A2.
Materials	The white root threads (or the added yarn) on the soil panel.
Steps	1. Finger starts in the seed pocket. 2. The learner follows one thread downward into the soil. 3. The finger then finds the stem path going up toward the leaves. 4. The learner shows a card or says the frame: "Root down. Shoot up."
Look for	The finger travels down into the soil and then up. The learner links that path to the sprout.

A4	Three Leaf Sizes
When	Touch station.
Materials	Loose leaves with 1, 2 and 3 raised veins. The matching leaves already on the model.
Steps	1. The learner feels the three loose leaves and orders them: one vein, then two, then three. 2. Match the small leaf to the seedling (3), the middle leaf to the adult plant (4), and the large leaf to the mature plant (5). 3. The learner checks by feeling the leaves glued on the model at those three places.
Look for	Leaves are ordered by touch and matched to stages 3, 4 and 5, with a reason such as "more leaf, older plant."

A5	Sun, Cloud and Soil by Touch
When	Guided practice.
Materials	Stiff sun rays, soft clouds, rough soil panel, smoother sky board. Four tactile need-cards: a small stiff sun, a cotton cloud, a sandpaper square, a card with a hole in it for air.
Steps	1. The learner finds the sun by its pointed rays, a cloud by its soft shape, and the soil by its rough panel. 2. Each find is matched to a need-card: sun to sunlight, cloud to water, sandpaper to soil. 3. For air, the learner holds the open card in the space beside the leaves and feels a gentle puff from a sheet of paper waved once. Say: “Air is here, around the leaves.”
Look for	Four needs are matched to four places on the model by touch.

A6	Five Body Heights
When	Recap. Children who prefer a small movement use one hand on the desk instead of the whole body.
Materials	Floor space, or a desk. The teacher calls the number, signs it, or taps the learner’s hand once for each stage.
Steps	1. Curl small for the seed (1). 2. One hand toward the floor for the root of the sprout (2). 3. Kneel or raise a hand to the waist for the seedling (3). 4. A hand at the shoulder for the adult plant (4). 5. A hand above the head for the mature plant (5), then the hand returns to the curled seed to show the cycle.
Look for	The learner moves through 1 to 5 in order and returns to the seed shape.

B. Visual activities (for deaf learners)

The teacher points, writes and shows. Instructions for these activities are on the board in short lines. Speech may happen, and the meaning is also on the card and on the model.

B1	Colour and Number Key
When	Development, while the model is uncovered. Leave the key on the board.
Materials	A key card and the board. Brown or orange = soil. Green = plant. Blue = sky. Gold = sun. Silver = cloud. Diamonds 1 to 5 = the stages.
Steps	1. Point to each colour on the real model and write the one word beside it. 2. Point to diamonds 1, 2, 3, 4 and 5 in order down the title panel. 3. A deaf learner places a matching colour chip or points to the matching word. 4. The key stays up for the rest of the lesson so later instructions can be a point, not a speech.
Look for	The learner can point to soil, plant, sky, sun and cloud, and can find diamonds 1 to 5.

B2	Diamond Order
When	Picture station in the group activity.
Materials	Five large diamond cards with the number, the stage word and a drawing.
Steps	1. The teacher points along the model from label 1 down to label 5, without a long explanation. 2. The learner places the five diamonds in the same order on the mat, left to right. 3. The learner checks each card against the white label on the model. 4. A child who is ready for three stages uses cards 1, 3 and 5 only.
Look for	Cards sit in the order seed, sprout, seedling, adult plant, mature plant.

B3	Same Plant, Five Heights
When	Picture station, after B2.
Materials	Five drawings made in the style of this model: seed under a soil line; a root and a tiny shoot; a short plant with two leaves; a taller plant; the tallest plant with roots showing.

Steps	1. Spread the drawings out of order. 2. The learner lines them up from the smallest plant to the tallest. 3. The learner points from each drawing to the matching place on the real model. 4. Ask, in writing or sign: “Which picture is the seedling? Which picture is the mature plant?”
Look for	The height order matches the model, and at least the seedling and the mature plant are named.

B4	Picture Needs on the Model
When	Picture station.
Materials	Four picture cards: sun, rain cloud, soil, air (curved arrows). Words are printed under the pictures in smaller type.
Steps	1. The learner places the sun card on the gold sun. 2. The cloud card goes on a silver cloud. 3. The soil card goes on the orange panel. 4. The air card is held in the blue space next to the leaves. 5. The teacher writes the four words in a row: sunlight, water, soil, air.
Look for	Each card sits on the matching part of the model.

B5	Arrow Back to the Seed
When	Recap.
Materials	A large curved arrow card. The real seed pocket. The mature plant on the right.
Steps	1. The learner puts the tail of the arrow on the mature plant and the head toward the seed pocket. 2. The board sentence is already written in large print: “The mature plant makes new seeds. The cycle starts again.” 3. The learner points: mature plant, then seed pocket, then label 1. 4. If there is time, the learner draws one seed dropping from the tall plant into the soil.
Look for	The arrow shows the return from Stage 5 to Stage 1, and the learner can point that path.

C. Sign language activities (for deaf learners)

Use the learner’s sign language. Where the class uses Pakistan Sign Language, use that, after the signs have been checked. The signer’s face stays in view. Hearing children copy the signs so the deaf learner is part of the class language.

C1	Sign on the Model
When	Development, at every stage stop. About one minute across the five stages, plus the need words.
Materials	The model, the word cards, and the agreed signs. Interpreter if the class has one.
Steps	1. At each stage do three things together: point to the part, show the written word, and sign the word. 2. Teach seed, water, sun, soil, root, leaf, stem, plant and grow with the real part in view. 3. Finger-spell sprout, seedling, mature and cycle, unless the learner already has a sign, and leave the spelling card on the table. 4. Children echo the sign. A deaf learner leads the echo when they wish.
Look for	The learner copies at least four signs while looking at the matching part of the model.

C2	Sign, Then Point
When	Sign station in the group activity.
Materials	Agreed signs and the model. Number diamonds as the answer cards.
Steps	1. The teacher signs one word and does not point. 2. The learner points to that part or holds up the number diamond. 3. Reverse the turn: the teacher points to a part, and the learner signs or finger-spells it. 4. Three turns each way are enough.

Look for	Three correct matches in each direction. A written card is a valid sign if the hand-sign is still new.

C3	Five Signs in a Line
When	Sign station, after C2.
Materials	Five children, five diamond cards, the model as the checker.
Steps	1. Children stand or sit in a line facing the class, hands and faces visible. 2. The first child signs Stage 1 and shows diamond 1. The line continues to Stage 5. 3. The group looks at the model labels and fixes any swap. 4. A hearing partner may sign with a deaf leader. The deaf learner chooses whether to lead.
Look for	The line runs from seed to mature plant, and each place has a sign, a finger-spelling, or the word card.

C4	Sign What the Plant Needs
When	Guided practice.
Materials	Signs for sun, water, soil and air. The four picture cards from B4.
Steps	1. The teacher signs one need. 2. The learner holds up the picture card and signs it back. 3. The learner then points to that part on the model: gold sun, silver cloud, orange soil, or blue space. 4. Repeat for all four needs.
Look for	Four needs are signed or matched, and each one is found on the model.

C5	Silent Story of the Cycle
When	Recap. This activity is done without spoken words.
Materials	The model and one partner. The curved arrow from B5.
Steps	1. A learner signs the five stages in order. 2. The partner moves a finger along the model in time with the signs, from label 1 to the tall plant. 3. Both finish by pointing from the mature plant back to the real seed pocket. 4. Classmates who are hearing use the same silence and the same signs.
Look for	The five stages are signed in order and the story returns to the seed.

Running the three stations. In the 10-minute activity, three tables work at the same time and then the groups change, or one group moves through all three tables if the class is small. Blind learners lead the touch table. Deaf learners lead the picture table and the sign table. Every child still does all three tables. The teacher stays mobile, with the face free for signing, and gives a hint only after the child has tried.

5. Procedure (40 minutes)

Previous knowledge to call on: children already know that a plant has roots, a stem and leaves, and that seeds are put in soil. This lesson joins those parts into one ordered cycle on their own model.

5.1 Initiation — The Closed Seed Box (5 minutes)

- Keep the model covered. Place only the closed seed box on the table.
- Open the box under a blind learner's hand and, at the same time, hold a seed up where deaf learners can see it clearly. Both groups meet the same object.
- Write on the board, in large letters: SEED. Under it, write beej in Urdu script by hand. Sign SEED. Say the word once, facing the class.
- Show two choice cards, YES and NO, for the written and signed question: "Will this seed stay a seed forever?" Collect answers by card, sign, speech or a nod.

- Uncover the model. Point to the pocket at label 1 and let one child feel that the seeds in the pocket match the seed from the box. Tell the class: “This board shows the life of this seed, from the pocket to the tall plant, and back again.”
- Ask for a prediction in any response mode: “How many stages are on the board?” Accept a number shown on fingers, a card, or a count along the diamonds.

5.2 Development — Walk the model from 1 to 5 (10 minutes)

Move from label 1 to label 5. At every stop, point, give the real or raised part to touch, show the word card, and sign the word (Activity C1). Keep each stop under two minutes. Use Activity A1 at Stage 1 and Activity B1 as you set the colour key.

- Stage 1, Seed. The real gram or bean is hard and dry. Inside a seed is a very young plant and a store of food, wrapped in a coat. On this model it rests in the soil. It is alive and it is waiting.
- Stage 2, Sprout. Give the class a careful look and a careful feel of the soaked bean (start of A2). Water enters the seed, the seed swells, and the coat opens. The first root, the radicle, grows down. The first shoot, the plumule, grows up. The white threads on the model are those roots.
- Stage 3, Seedling. The shoot reaches the air and the light. The stem is thin. The first small leaves open. Show the small paper leaves on the model and the one-vein loose leaf.
- Stage 4, Adult plant. The stem is longer and there are more leaves. Roots hold the plant and take in water from the soil. The leaves use sunlight, water and air (carbon dioxide) to make the plant’s food.
- Stage 5, Mature plant. This is the tallest plant on the board, with the largest leaves and roots still showing. A mature plant can make flowers and then new seeds. Those seeds can fall back into the soil. That return is the cycle: Stage 5 leads to Stage 1. Point from the tall plant back to the real seed pocket.
- Needs, shown on this model. Gold sun = sunlight. Silver clouds = water. Orange panel = soil. Blue space around the leaves = air. Sign each need as you point to it.
- One local link, said in a single sentence: the gram we cook is a seed, and a gram plant goes through these same five stages.

5.3 Activity — Three stations on the model (10 minutes)

This period runs the activity section. Set three stations before the lesson starts, close enough that you can see every child and far enough apart that each group has room.

- Touch station (multisensory): Activities A2, A3 and A4. Dry seed and soaked seed, the white root, and the three leaf sizes. A blind learner leads. Others touch the same objects. Seeds return to the lid or the pocket before the group moves.
- Picture station (visual): Activities B2, B3 and B4. Diamond cards in order, the five height drawings, and the four need cards placed on the model or on a photo of the model if the real model is in use at another station. A deaf learner leads. The instruction card on the table reads: “1. Order the diamonds. 2. Order the heights. 3. Place sun, cloud, soil, air.”
- Sign station (sign language): Activities C2 and C3. Sign then point, then five signs in a line. The signer faces the group. Hearing children copy the signs.
- Minutes 0 to 1: show the routine once. Touch one seed, place one diamond, sign one word.
- Minutes 1 to 4, 4 to 7, and 7 to 10: each group works at one station, then moves. With a very small class, one group walks the three stations in the same order.
- The teacher watches who leads, who gets a turn with the real seed, and whether every deaf learner can see the signer. Feedback is specific: “You found the root going down,” or “Your diamond 3 matches the seedling.”

5.4 Guided practice and assessment — One task each (10 minutes)

Give each child one task at a fitting level. Record it on the checklist in Section 6. Extra wait time is part of the task, and hints are reduced once the child starts. A signed, pointed, matched or made answer counts fully.

- Task A, remember. Point to the stage the teacher names or signs, or match one label to its place on the model.
- Task B, understand. Place the dry seed at Stage 1 and the soaked seed at Stage 2, or match the three leaf sizes to stages 3, 4 and 5 (Activities A2 and A4).
- Task C, apply. Put diamonds 1 to 5 in order, or match the four needs by touch or by picture (Activities A5, B2, B4 and C4).
- Task D, analyse. Compare the seedling and the mature plant. Ask: "What is the same? What is different?" Accept speech, sign, two cards, or a hand moving from the small leaves to the large leaves.
- Task E, extend. "Why does the arrow go back to the seed?" or "The sun on the model is covered. What happens to the adult plant, and why?" A strong answer names food-making in the leaves, or names the missing need.

5.5 Recapitulation — Back to the pocket (5 minutes)

- Activity B5: one child places the curved arrow from the mature plant to the seed pocket.
- Activity C5: one pair tells the silent signed story while a finger travels along the model.
- Activity A6: the class makes the five body heights and returns to the seed shape. Children who want a smaller movement do it with one hand on the desk. Call the stages by number, by sign, and by pointing, so blind and deaf learners share the same sequence.
- Exit check, any response mode: "Which stage is the real seed?" "Which stage has the first small leaves?" "Which stage can make new seeds?" "Name one thing the plant needs and show it on the model."
- Closing, written in large print, signed, and said once while facing the class: "The seed in our pocket can become the tall plant on this model. It sprouts, becomes a seedling, grows into an adult plant, and becomes a mature plant that makes new seeds. Sunlight, water, air and soil are what it needs on the way."
- Seeds are counted back into the box. Hands are washed. The model is left where the next class can find it in the same orientation.

6. Individual Assessment Checklist

Mark the checklist during Section 5.4. I = the child did it alone. P = the child did it with a hint or a smaller set of stages. A sign, a point, a match or a demonstration is marked the same way as speech.

Student	Names 5 stages	Orders 1 to 5	Seed or a need	Cycle or comparison	Notes
	☐ I ☐ P	☐ I ☐ P	☐ I ☐ P	☐ I ☐ P	
	☐ I ☐ P	☐ I ☐ P	☐ I ☐ P	☐ I ☐ P	
	☐ I ☐ P	☐ I ☐ P	☐ I ☐ P	☐ I ☐ P	
	☐ I ☐ P	☐ I ☐ P	☐ I ☐ P	☐ I ☐ P	
	☐ I ☐ P	☐ I ☐ P	☐ I ☐ P	☐ I ☐ P	

Notes can record the access route that worked: real seed, picture cards, signs, fewer stages, extra time, partner support.

7. Inclusive Accommodations and Differentiation

Learners	How they take part in this lesson	How learning is shown
Blind learners and learners with low vision	Real seeds in the pocket, the soaked bean, yarn roots, three leaf sizes, stiff sun and soft clouds. Spoken position of the model. Raised or Braille numbers. The model is never moved mid-lesson. Glare is kept off the board for learners who have some sight. Large print is offered as well as touch.	Finding the seed, following the root, ordering the leaves, and explaining in speech or by placing objects. Pictures and colour are an extra, and the task can be finished without them.

Learners	How they take part in this lesson	How learning is shown
Deaf learners and learners with hearing loss	Visual method and sign language together: colour-and-number key, written steps at each station, demonstration, and the child's sign language with the face visible. An interpreter signs if one is part of the class. Background noise is lowered so children who use hearing aids can use them. The teacher writes, then turns to face the class.	Signing, finger-spelling, pointing, matching cards, writing, or moving the arrow. Listening is a support, and the science mark does not depend on it.
Deafblind learners	One object at a time: real seed, then yarn root, then one leaf. Agree a touch cue for yes, no and next before the activity. Hand-under-hand guidance is used when the learner wants it. Stages 1, 3 and 5 may be enough for this period.	A reliable touch response, an object placed on the right stage, or a touch sign the team already uses.
Learners who speak little or use another communication method	YES/NO cards, picture choices, pointing, partner-assisted scanning, or a device they already use. Wait after each question.	A selected card, a point, or a placed object is a full answer.
Learners who need a shorter step or a quieter space	One instruction at a time. Model the step, then hand over. Offer stages 1, 3 and 5 before adding 2 and 4. A short movement break can be Activity A6 done early.	Mark the stages the child actually secured. Speed is recorded in the notes, and it is separate from the science mark.
Learners with limited movement	The model, the seed box and the cards are placed where the child's hands or pointer already work. Large pieces and a non-slip mat. A partner moves an item when the child directs it.	Eye gaze, a switch, speech, or a directed point counts. The child decides; the partner only carries out the move.
Learners who are ready for more	Same model and same stations. Ask for the words radicle and plumule at Stage 2, for a reason about food-making in the leaves, and for a local plant (gram, wheat or mango) mapped onto the five stages.	An accurate sequence, a clear comparison, and a reason that uses the model.

8. Board Summary

Build this table on the board during the development, one row at a time. Leave it up until the closing sentence.

Stage on our model	What the child can find	Science idea to leave on the board
1 Seed (beej)	Real gram and bean seeds in the soil pocket	A young plant and stored food, waiting in the soil
2 Sprout (ankur)	White root threads; soaked bean in the lid	Water opens the seed. Root grows down. Shoot grows up.
3 Seedling (nau-khez pauda)	Short stem and the small leaves	First leaves open in the light
4 Adult plant (bara pauda)	Taller stem and more leaves	Leaves make food with sunlight, water and air
5 Mature plant (baligh pauda)	Tallest plant, large leaves, roots showing	Makes new seeds. The cycle returns to Stage 1

Needs line, copied under the table: Sunlight (gold sun) · Water (silver clouds) · Soil (orange panel) · Air (blue space by the leaves).

Cycle line: Mature plant → new seeds → seed pocket → the five stages begin again.

9. Assessment Criteria and Success Indicators

- **Beginning.** the learner shows at least one stage with the model, a hint, or a choice card.
- **Developing.** the learner identifies most of the stages and matches at least three labels, seeds or leaves to the model.
- **Achieving.** the learner identifies all five stages and places them in order from seed to mature plant.
- **Extending.** the learner compares two stages, names what the plant needs, and gives a reason for the cycle or for a covered sun / a dry seed.

A response in sign, touch, print, picture or speech can fall in any of these four bands. The band describes the science, and the notes describe the support.

10. Reflection and Follow-up

After the class, the teacher writes four short notes:

- Which part of the model worked by touch: the real seeds, the yarn, the leaves, the sun or the clouds.
- Whether every blind learner could find the model again, and whether every deaf learner could see both the signer and the board.
- Which stage was confused most often, especially sprout versus seedling, or adult versus mature.
- Who needs the three-stage set again, and who is ready to map a gram plant or a mango tree onto the same cycle.

Follow-up in the next science period: plant the extra gram seeds in a clear cup so the class can watch a real root and shoot over the week, and match the cup to Stages 1, 2 and 3 on this model. Children who do not handle soil can observe the cup and record the change on a picture strip.

Prospective Teacher's Signature	Supervisor's Signature
Hassnain Signature _____ Date _____	Signature _____ Remarks _____ Date _____

End of lesson plan